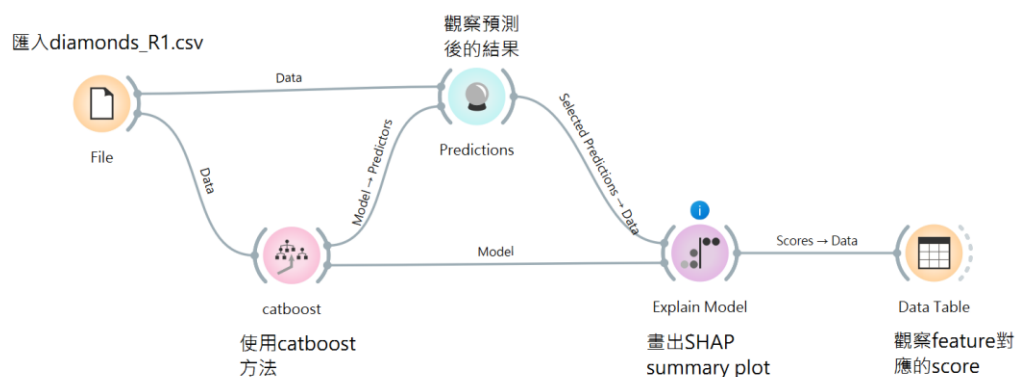


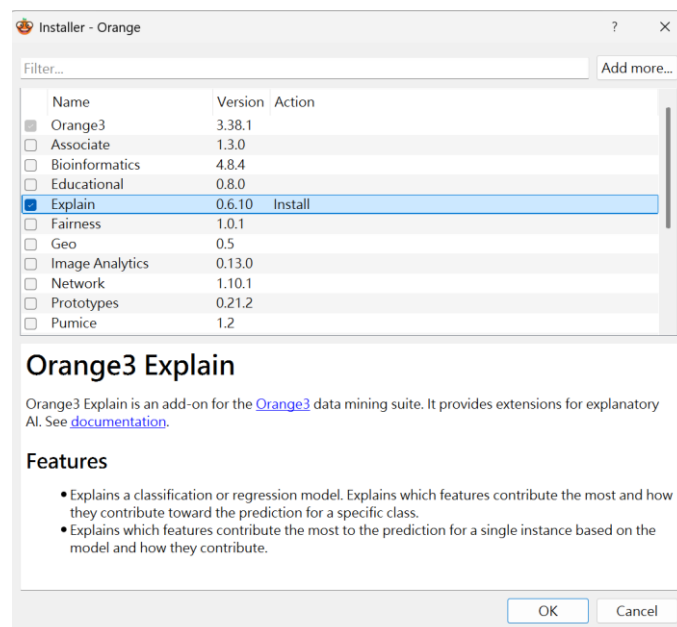
流程圖一

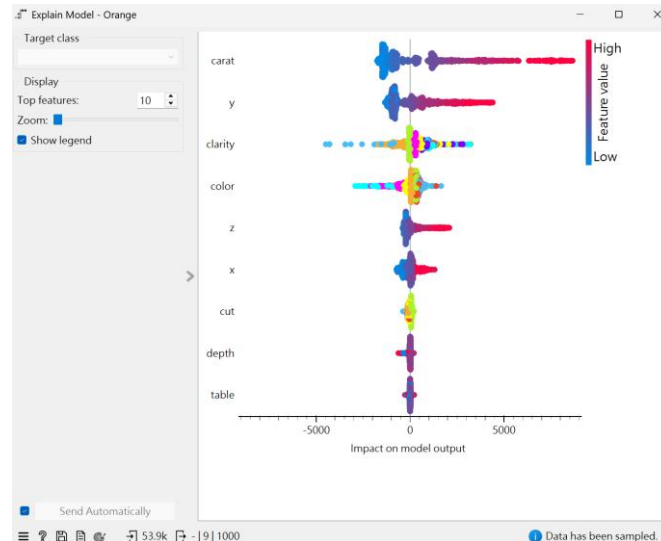
流程圖一



1. 打開 diamonds_R1.csv，使用 catboost 與 predictions 建立 SHAP summary plot (10%)

由於 orange 本身內建沒有 explain 的模組，因此先到 installer 中下載 Explain 模組。接著連接 catboost, prediction，最後連接 explain model 畫出 SHAP summary plot 進行觀察。觀察後可以發現屬性重要度排序為 carat > y > clarity > color > z > x > cut > depth > table

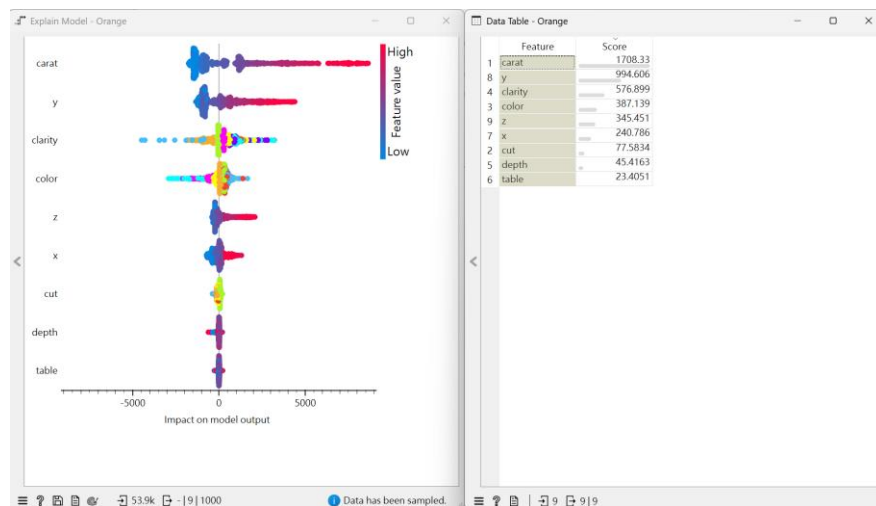




2. SHAP summary plot 所呈現的「特徵重要性排序」合理嗎？為什麼？(10%)

我認為結果合理。因為 target 是 price，按照常理來說 carat 應該與 price 非常有關，其次是和色澤相關的屬性，鑽石的形狀則較沒有影響。

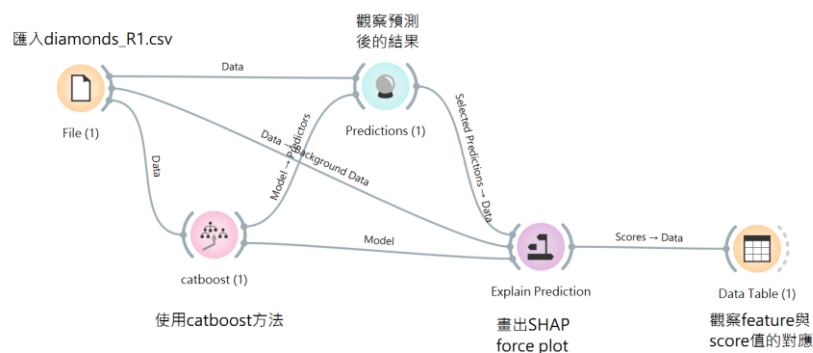
3. 就 carat 而言，說明屬性值與 shap value 的關係。(建議利用 Explain Model 的輸出值。)(20%)



屬性的 Shap value 越高，該屬性對於 target 的預測影響越大。

流程圖二

流程圖二



1. 繪出以下 instance 的 SHAP force plot 並解釋如何算出預測值。(20%)

	Gradient Boosting (1)	error	price	carat	cut	color	clarity	depth	table	x	y	z
14	11469	0	9823	3.11	Fair	J	I1	65.9	57.0	9.15	9.02	5.98

在 prediction widget 中選出題目給的 instance，並利用 explain prediction 畫出 shap force plot。預測值計算方式如下：

Base value: 3183.2

各屬性值對 base value 的影響(從 shap force plot 得出此數據)：

Carat: 增加 7410.0

Y: 增加 2940.42

Z: 增加 1556.99

Color: 增加 340.72

Table: 減少 96.39

Depth: 減少 219.24

X: 減少 244.02

Cut: 減少 727.01

Clarity: 減少 2989.94

預測值： $3183.2 + 7410.0 + 2940.42 + 1556.99 + 340.72 - 96.39 - 219.24 - 244.02 - 727.01 - 2989.94 = 11154.73$

Predictions (1) - Orange

Shown regression error: Difference

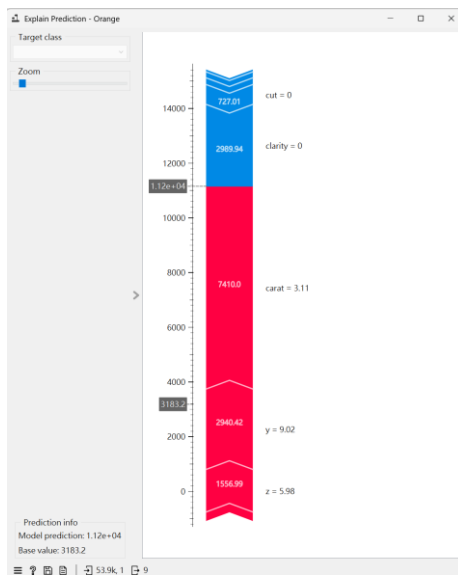
Restore Original Order

	catboost (1)	error	price	carat	cut	color	clarity	depth	table	x	y	z
5373	10456	615	9841	1.51	Very Good	G	SI1	62.9	58.0	7.25	7.28	4.57
5374	9607	-230	9837	1.30	Very Good	G	VS1	62.5	57.0	6.92	6.96	4.34
5375	9068	-768	9836	1.64	Premium	J	VS1	60.6	59.0	7.67	7.61	4.63
5376	9455	-381	9836	1.23	Premium	D	VS2	59.3	59.0	7.03	6.96	4.15
5377	9735	-101	9836	1.23	Ideal	D	VS2	62.3	57.0	6.86	6.82	4.26
5378	9818	-18	9836	1.23	Ideal	F	VS1	62.5	54.0	6.85	6.82	4.27
5379	10428	595	9833	1.51	Ideal	H	SI1	61.3	56.0	7.40	7.44	4.55
5380	8938	-895	9833	1.51	Good	E	SI2	63.2	56.0	7.26	7.32	4.61
5381	9489	-344	9833	1.51	Premium	H	SI1	61.3	60.0	7.29	7.33	4.48
5382	8989	-842	9831	1.52	Premium	E	SI2	58.1	60.0	7.59	7.53	4.39
5383	9314	-517	9831	1.08	Ideal	G	IF	62.3	56.0	6.55	6.59	4.09
5384	9795	-33	9828	1.50	Very Good	I	VVS2	63.3	58.0	7.24	7.21	4.57
5385	8983	-845	9828	1.50	Good	I	VS1	57.9	60.0	7.48	7.44	4.32
5386	11893	2065	9828	1.50	Ideal	E	SI1	61.9	57.0	7.37	7.31	4.54
5387	9633	-195	9828	1.50	Premium	I	VS1	61.6	59.0	7.32	7.26	4.49
5388	9142	-685	9827	1.50	Ideal	I	VS2	60.0	61.0	7.39	7.42	4.44
5389	9945	122	9823	1.53	Very Good	I	VS1	59.3	58.0	7.50	7.54	4.46
5390	11155	1332	9823	3.11	Fair	J	I1	65.9	57.0	9.15	9.02	5.98
5391	9755	-66	9821	1.51	Good	H	VS2	64.0	59.0	7.19	7.25	4.62
5392	10695	874	9821	1.51	Very Good	H	VS2	62.3	61.0	7.27	7.30	4.54
5393	9656	-165	9821	1.54	Premium	D	SI2	59.9	62.0	7.51	7.44	4.48
5394	9523	-297	9820	1.12	Very Good	G	IF	61.0	58.0	6.70	6.72	4.09
5395	9045	-775	9820	1.12	Ideal	G	VVS2	61.6	55.0	6.68	6.74	4.13
5396	9417	-403	9820	1.50	Premium	H	SI1	61.0	60.0	7.33	7.27	4.45
5397	9399	-420	9819	1.63	Ideal	J	VS1	61.8	56.0	7.53	7.58	4.67
5398	10517	699	9818	1.55	Very Good	H	SI1	61.6	54.0	7.41	7.47	4.58
5399	8810	-1007	9817	1.50	Very Good	F	SI2	62.9	55.0	7.26	7.28	4.57
5400	8800	-1017	9817	1.52	Very Good	D	SI2	58.6	59.0	7.43	7.48	4.37
5401	10203	386	9817	1.10	Ideal	F	VVS2	61.2	56.0	6.68	6.65	4.08

Show performance scores

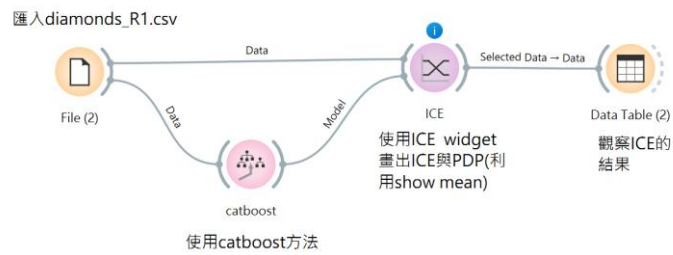
Model	MSE	RMSE	MAE	MAPE	R2
catboost (1)	269267.019	518.909	290.819	0.088	0.983

53.9k | 1x53932



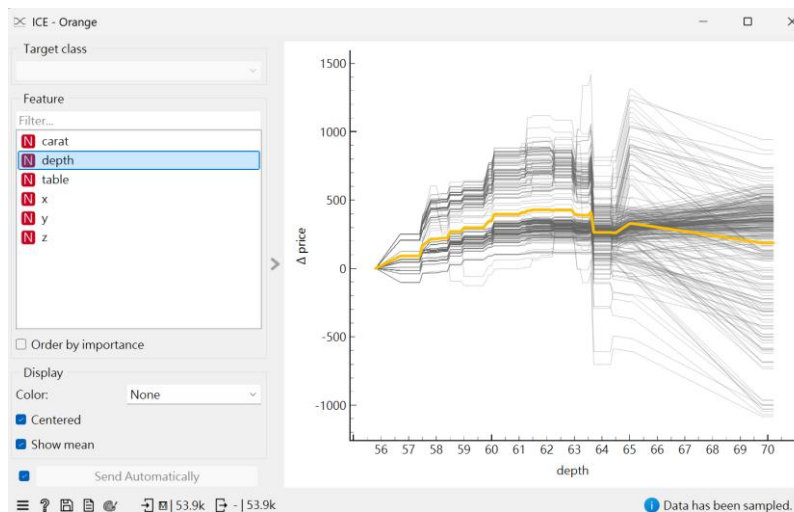
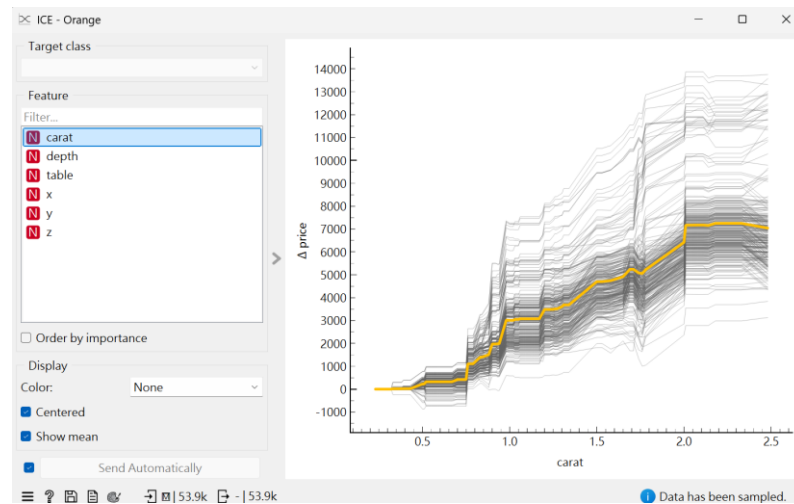
流程圖三

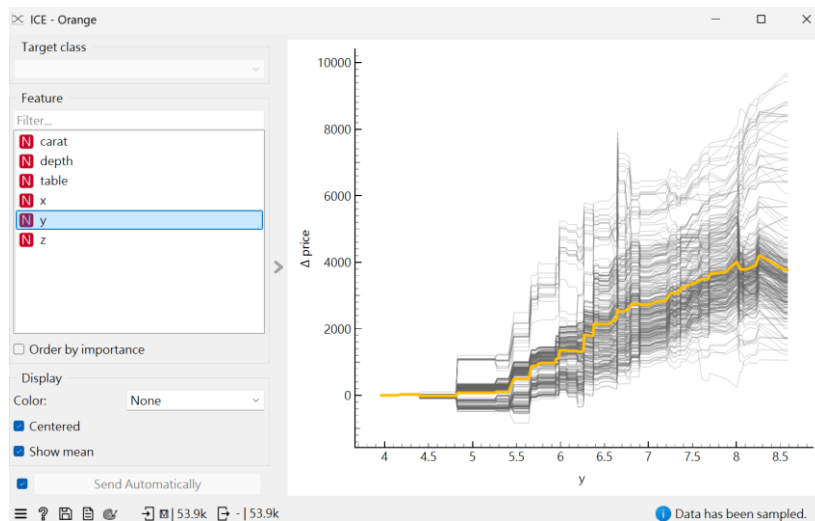
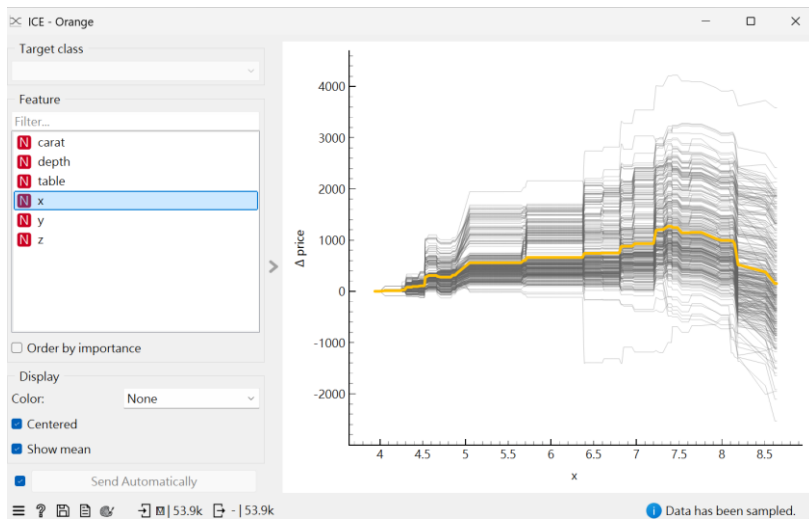
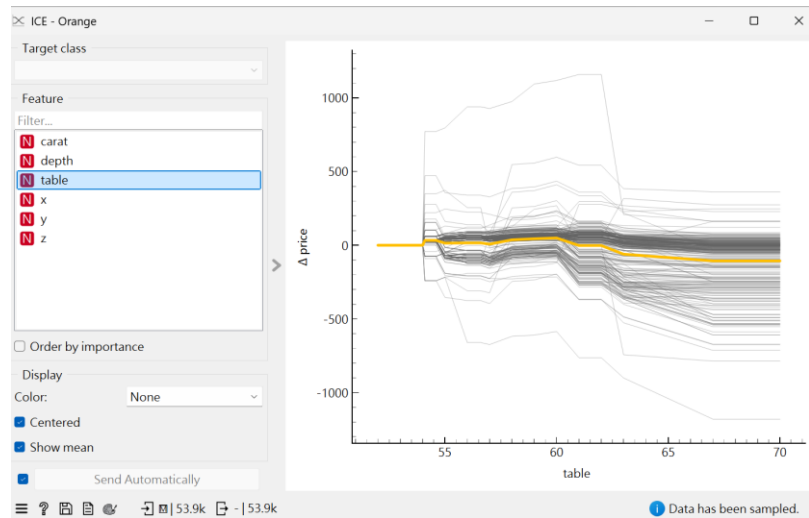
流程圖三

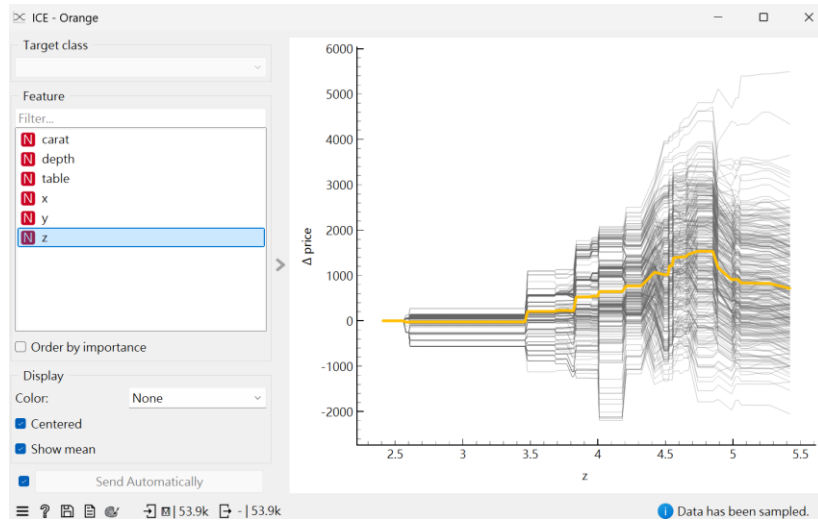


1. 打開 diamonds_R1.csv，使用 catboost，繪出所有 numeric 屬性的 PDP (Partial Dependence Plot)，是否可以使用這些圖來解釋屬性的重要性排序？(20%)

這題我連接 ICE widget 並使用 show mean 畫出黃線幫助我評估。







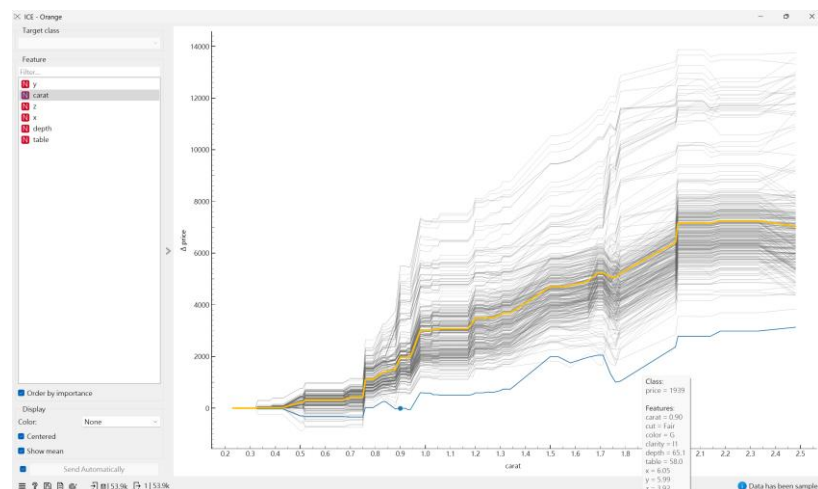
黃線代表 mean，如果黃線是水平線，代表該屬性對 target 沒有影響，斜率的絕對值大代表，對於 target 的影響越大，所以根據上面的圖，屬性重要度的排序是：carat > y > z > x > depth > table

2. 就每一個屬性，找出 ICE 與 PDP 不同的一個或多個 instance(s)。(20%)

本題我就每一個屬性在 ICE 中圈選與黃線不同的一個 instance。

Carat:

Data Table (2) - Orange										
Info										
1 instance (no missing data)										
9 features										
Numeric outcome										
No meta attributes.										
Variables										
Show variable labels (if present)										
Visualize numeric values										
Color by instance classes										
Selection										
Select full rows										
price	carat	cut	color	clarity	depth	table	x	y	z	
1939	0.90	Fair	G	I1	65.1	58.0	6.05	5.99	3.92	



Y:

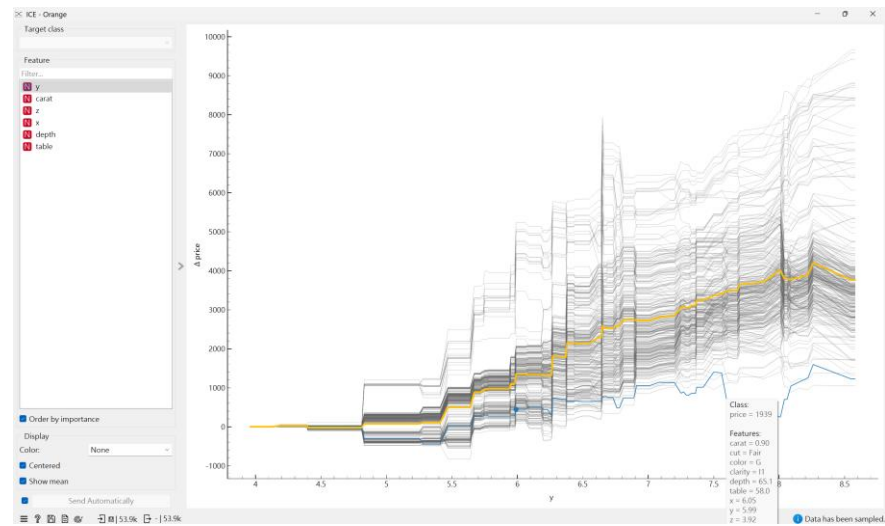
Data Table (2) - Orange

Info
1 instance (no missing data)
9 features
Numeric outcome
No meta attributes.

Variables
☒ Show variable labels (if present)
☒ Visualize numeric values
☒ Color by instance classes

Selection
☒ Select full rows

	price	carat	cut	color	clarity	depth	table	x	y	z
1	1939	0.90	Fair	G	I1	65.1	58.0	6.05	5.99	3.92



Z:

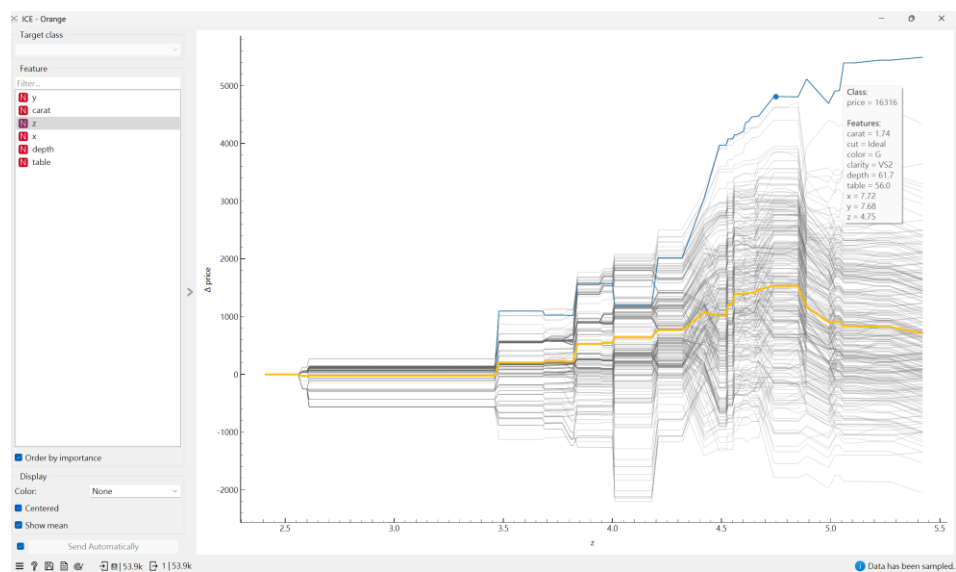
Data Table (2) - Orange

Info
1 instance (no missing data)
9 features
Numeric outcome
No meta attributes.

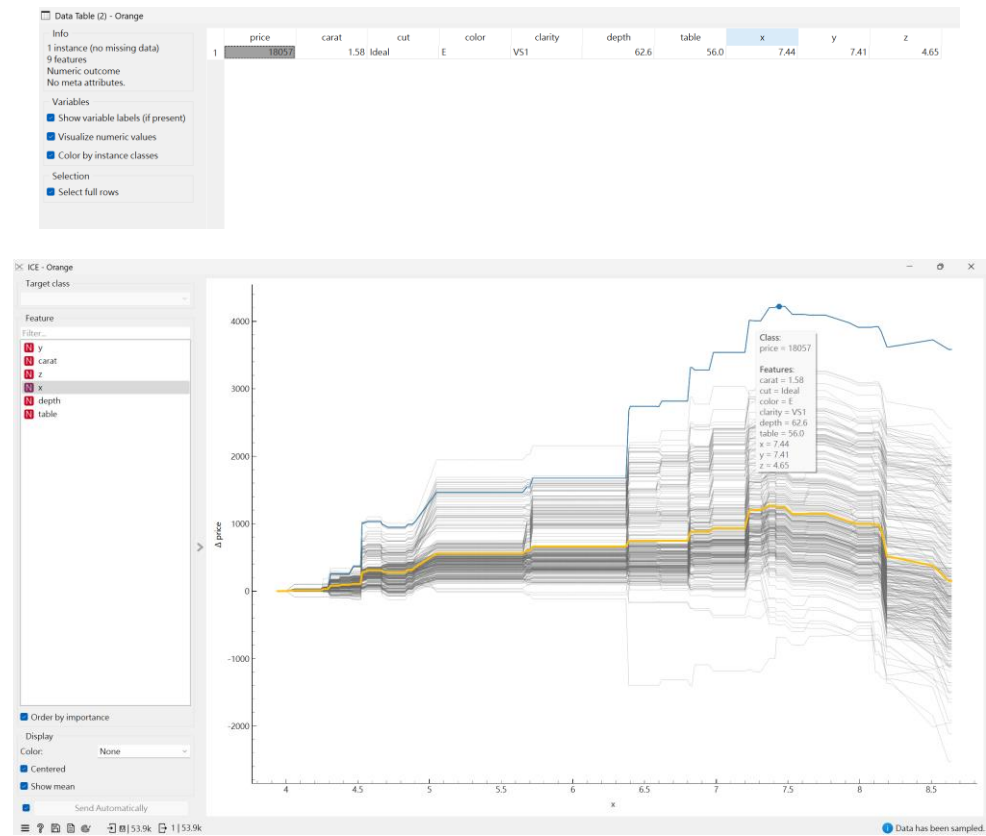
Variables
☒ Show variable labels (if present)
☒ Visualize numeric values
☒ Color by instance classes

Selection
☒ Select full rows

	price	carat	cut	color	clarity	depth	table	x	y	z
1	16316	1.74	Ideal	G	VS2	61.7	56.0	7.72	7.68	4.75



X:



Depth:

Data Table (2) - Orange

Info
1 instance (no missing data)
9 features
Numeric outcome
No meta attributes.

Variables
☒ Show variable labels (if present)
☒ Visualize numeric values
☒ Color by instance classes

Selection
☒ Select full rows

	price	carat	cut	color	clarity	depth	table	x	y	z
1	8220	2.33	Premium	H	I1	60.7	58.0	8.64	8.56	5.22

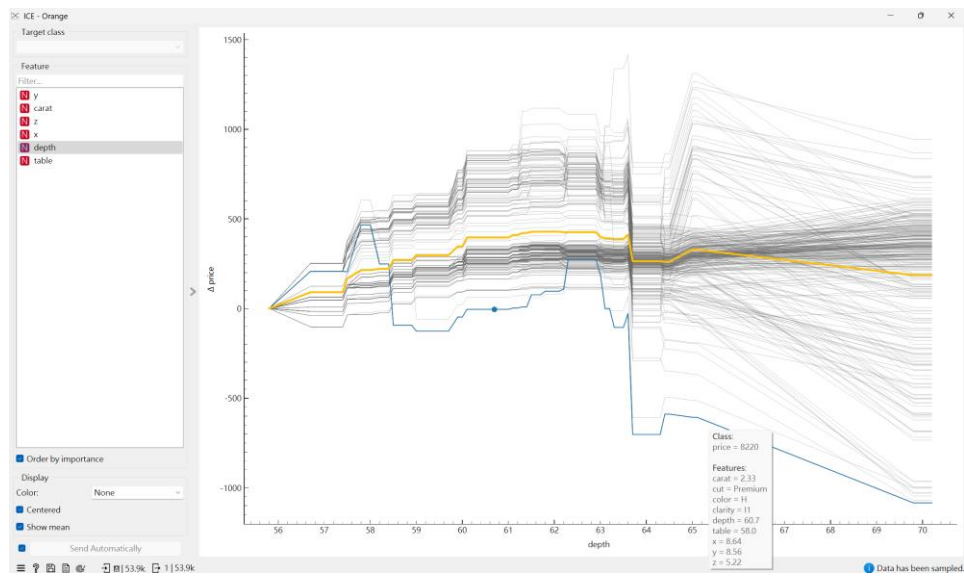


Table:

Data Table (2) - Orange

Info

1 instance (no missing data)

9 features

Numeric outcome

No meta attributes.

Variables

- Show variable labels (if present)
- Visualize numeric values
- Color by instance classes

Selection

- Select full rows

	price	carat	cut	color	clarity	depth	table	x	y	z
1	16336	2.11	Premium	G	SI2	62.2	60.0	8.19	8.15	5.08

